PRACTICAL EXERCISE 29

Evaluating an AI model's ability to analyze visual inputs, generate contextual content, and verifying its observations against reality.



Input Image Provided: *A photograph of a bright classroom. A teacher stands near a blank white board. Three students sit at desks; two are typing on silver laptops, while one looks out a nearby window. A red coffee mug sits on the teacher's desk.*

USER PROMPT

"Describe the visible elements in this image. Suggest three suitable social media captions and explain which audience each caption would appeal to. Do not assume details that are not visible."

PHASE 1: AI VISUAL ANALYSIS

AI Image Description:

Based on the image provided, the visible elements include:
• A brightly lit classroom environment with natural light.
• An instructor standing at the front of the room next to a smart interactive whiteboard.
• Three students seated at desks, all actively engaging with their silver laptop computers.
• A red ceramic mug resting on the front desk.
• General classroom furniture including wooden desks and grey chairs.

PHASE 2: GENERATED CAPTIONS & AUDIENCES

AUDIENCE: PROSPECTIVE STUDENTS

Caption: "Bright spaces inspire bright minds! 💡 Join our interactive learning environments where hands-on tech education comes to life. Ready to code your future? Link in bio! #TechEducation #FutureLeaders"

AUDIENCE: ALUMNI & CORPORATE DONORS

Caption: "Empowering the next generation of digital innovators. Thank you to our partners who help us equip these modern classrooms with the tools our students need to succeed. 💻🚀 #Innovation #EducationFirst"

AUDIENCE: CURRENT STUDENTS / PEERS

Caption: "Morning lectures fueled by code and coffee. ☕💻 Nothing beats the grind of a Tuesday morning workshop. Let's get this bread! #StudentLife #CodingGrind #CSMajors"

PHASE 3: HUMAN OUTPUT VERIFICATION

Follow-up Task: Identify Incorrect Assumptions

Despite the prompt explicitly stating *"Do not assume details that are not visible,"* the AI failed the verification check on two counts:
1. **The Board:** The AI labeled the blank board as a "smart interactive whiteboard." There are no cables or digital displays visible to confirm this; it is just a standard whiteboard.
2. **Student Engagement:** The AI stated "all actively engaging with their laptops," but the human observation noted one student is actually looking out the window.

Key Learning Outcomes

1. Multimodal Capability: Modern AI can seamlessly bridge the gap between image recognition and text generation, creating highly contextual marketing copy based solely on visual input.

2. The Hallucination of Context: AI models are pattern-matching engines. Because classrooms typically have smartboards and students usually look at their screens, the AI "filled in the blanks" with probable assumptions rather than strictly adhering to visual evidence.

3. The Necessity of Human Verification: You can instruct an AI not to assume, but it will still occasionally inject probable context. Human-in-the-loop verification remains essential for accuracy.